

Grand Quize Mega File

BIO301

MCQS

1. Primary oocyte produces polar bodies ……

(a) 1

(b)2

(c)3

(d)4

2. Which two alleles are called co-dominant one another and are expressed when present?

(a) Allele A and Allele B

(b) Allele A and Allele O

(c) Allele B and Allele O

(d) Allele A and Allele i

3. During recessive epistasis ,which color of pigeon is produced by recessive red gene with or without spread?

(a) Ash red

(b) Blue

(c) brown

(d) Any color

4: The DNA between nucleosome is called as linker DNA .Its length is …….

(a)20-80 bp

(b) 10-20 bp

(c)20 -40 bp

(d) 20-60 bp

5:Mendel determined the segregation of alleles at single gene locus based on …..

(a) Monohybrid

(b) Dihybrid

(c) Trihybrid

(d) All of these

6: Darker bands are generally found near the centromere or on the ends (telomeres) of chromosomes.The position of the dark –staining are …….or……..

(a)Heterochromatic region ,heterochromatin

(b)Euchromatin region,euchromatin

(c) Euchromatin region, heterochromatin

(d)Heterochromatin region, euchromatin

7:During metaphase mitosis chromosome

(a) Undergo coiling

(b) Line up at the equator

(c)Chromosomes condense and become visible

(d)Break the nuclear envelope

8:During stage telophase 1 ,chromosomes are……

(a) Haploid
(b) Diploid
(c) Haploid and Diploid
(d) Polyploid

9:The law of independent assortment with dihybrid cross gives genotypic ratio in F2 as:

(a) 1:1
(b) 3:1
(c) 1:2:1
(d) 9:3:3:1

10:Who is considered "The Father of Genetics"?

(a) Charles Darwin
(b) Gregor Johann Mendel
(c) James Watson
(d) Francis Crick

11:In …….centromere splits into two parts while in ……it does not split.

(a)Mitosis,Meiosis I

(b)Meiosis I ,Mitosis

(c)Mitosis,Meiosis II

(d)Meiosis II,Mitosis

12:When one phenotypic trait is determined by multiple alleles then it is called as ……

(a) Epistasis

(b) Pleiotropic

(c) Polygenic inheritance

(d) All

13:How many numbers of genes are present in drosophila?

(a) 5800
(b) 10000

(c) 13700

(d) 4500

14:Which is hereditary disorder characterized by lung congestion ,infection,and malabsorption of nutrients by the pancreas?

(a) Sickle cell anemia
(b) Cystic fibrosis
(c) Huntington disease
(d) Alzaheimer 's disease

15:which one of the following histone protein has molecular weight ,greater number of amino acid and high percentage of lysine and arginine?

(a)H1 (b) H2A (c)H3 (d) H4

16 ..... is an example of inherited variation?

(a) Language (c) Atheletic skill
(b) Obesty (d) Finger prints

17:When chromosomes stained they show a dark –stained region that is ……..

(a) Telomere
(b) Centromere
(c) Chromatids
(d) DNA

18:Histones have two basic amino acids ……

(a) Valine –lysine

(b) Arginine-lysine

(c)Arginine –methionine

(d)Tyrosine-valine

19:The diameter of DNA nuclosome on the chromosome scaffold is ?

(a) 2nm
(b) 11nm
(c) 30nm
(d) 300nm

20:E.coli has about ..... genes are organized into its one circular chromosome

(a)1000 to 2000

(b)2000 to 3000

(c)3000 to 4000

(d)4000 to 5000

21:Which is the following blood group is considered a universal donor?

(a) A (c)AB

(b) B (d)O

22:Which type of genetics deals with movement of genes and genetic traits from parents to offspring?

(a)Population genetics

(b)quantitative genetic

(c)Molecular genetic

(d)Transmission genetics

23:Which statement is correct about genetic and physical mapping?

(a) The some genes are far apart on the genetic map,but close together on the physical map of the chromosome

(b) The some genes are far apart on the genetic map and also are far apart on the physical map of the chromosomes

(c) The some genes are close together on the genetic map and are also close together on physical map of the chromosome

(d) all of these

24:Polygenic traits are also affected by the persons environment .so they are called

(a) Mulltifactorial
(b) Environmental influence
(c) Major factor
(d) Environmental effects

25:Which is the histone deacetylase inhibitor?

(a)Vorinostat (d) all

(b)Lapatnib

(c)Sunitinib

25:Generally,karyotype represent by arranging in the …….order of size keeping their centromere in ……?

(a) Ascending ,vertical line

(b) Descending ,straight line

(c)Ascending ,straight line

(d)descending,vertical line

26:The spindle apparatus is formed during the ……phase of mitosis.

(a)Prophase

(b)Anaphase

(c)Metaphase

(d)Telophase

27:During ……the effect of prophase and pro metaphaseare reversed.

(a)Interphase

(b)Anaphase

(c)Metaphase

(d)Telophase

28:Meiosis result in…..

(a)2 haploid daughter cells

(b)4 diploid cells

(c) 2 diploid cells

(d) 4 Haploid daughter cells

29:The allelic series refer to

(a)Multiple alleles

(b)Dominant alleles

(c)Recessive alleles

(d)Lethal alleles

30:Which of the following statement is correct about genes

(a)30% are mono allelic and they exist in two forms

(b)30 % are mono allelic and they exist in only one form

(c)70% are di allelic

(d)70% are mono allelic and they exist in one form

31:The crossing over occurs at

(a)Desmosomes

(b)kinetochore

(c)Chiasmata

(d)Centromeres

32:The longest stage in the cell cycle is

(a)Interphase

(b)Anaphase

(c)Metaphase

(d)Telophase

33:Which of the following statement is not correct about continuous variation?

(a)Continuous variation controlled by a lot of genes

(b)Continuous variation strongly influenced by the environment

(c)Continuous variation has distinct categories

(d)Continuous variation graph to fit a smooth curve

34 ..... have the same length,shape,centromere location,and genes

(a)Paired chromosomes

(b)Autosomes

(c)Sex chromosomes

(d)XY chromosomes

35:Mutational changes are …….and epigenetic changes can be……

(a)Reversible,reversible

(b)Irreversible,irreversible

(c)Reversible,irreversible

(d)Irreversible,reversible

36:In …….sister chromatids separate and move along the kinetochore microtubules toward opposite ends of the cell.

(a) Anaphase

(b) Metaphase
(c)Telophase
(d)Prophase

37:Position dependent method of diseased gene identification follows

(a) Forward genetics (d)Relative genetics

(b) Reverse genetics

(c)Simple genetics

38:Which statement is not correct about epigenetic regulation in identical twins?

(a) The difference in the twins epigenomes is what makes them become different when they are older

(b) Over time the twins environment will diverge ,resulting in individual epigenetic tags to form for each twins

(c) When infants they experience the same or very similar environmrnt.so there is little variation in epigenome

(d) The epigenetic tags can have such an effect on the twins that one can develop a disease and the other is also

39:Which of the following genotype expresses the Green color of pepper?

(a)RR\YY

(b)Rr|\yy

(C)rr\Yy

(d)rr\yy

40:Which one of the following statement is correct?

(a) Father with X-linked disorder will transmit the mutant allele to all sons but to no daughters

(b) Father with X-linked disorder will transmit the mutant allele to all daughters but to no sons

(c) Father with X-linked disorder will transmit the mutant allele to half daughters and half to sons

(d) Father with X-linked disorder will transmit the mutant allele to half daughters and all to sons

41:Band that indicates constitutive heterochromatin……

(a) G band
(b) Q band
(c) C band
(d) R band

42:What is the process of transfer of human blood known as?

(a) Transforming

(b) Transfusion

(c)Processing

(d)Transporting

43:All offspring of same variety is called as

(a) Cross breed

(b)Hybrid

(c)Line breed

(d)True breed

44:In DNA double helix,the two DNA chains are held together by

(a) Covalent bonds between the pair of bases
(b) Hydrogen bonds between the pair of bases
(c) Ionic bonds between the pair of bases
(d) None

45:If 14% of the nucleotides from a DNA molecules contain the base T.what percent will contain the base G?

(a)14% (b)18% (c)28% (d)36%

46 ……. are proteins used to package DNA

(a)Telomerases

(b)Chromatins

(c)Nucleosomes

(d) Histones

47:Darker bands are generally found near the centromere or on the ends(telomeres)of the chromosome.The position of the dark staining are_____

(a) Heterochromatin region,heterochromatin
(b) Euchromatin region,euchromatin
(c) Euchromatin regin,heterochromatin
(d) Heterochromatin region ,euchromatin
(e)

48:According to……. genetic material of parents mixes during fertilization.

(a) Particulate hypothesis

(b) Blending hypothesis

(c)both

(d) none

49:Epistasis is the interaction between……genes.

(a) 2
(b) 4
(c) 6
(d) 8

50:On what basis four major blood groups is classified as positive and negative blood type?

(a) Antigen
(b) Antibody
(c) Rh factor
(d) RBCs

51:Which one of the following statement is correct about to the Mendel Law of Dominance?

(a) In a cross of parents that are pure for contrasting traits only one form of the trait will appear in the next generation

(b) In a cross of parents that are pure for contrasting traits blended form of the trait will appear in the next generation

(c) In a cross of parents that are pure for contrasting traits two form of the trait will appear in the next generation

(d) In a cross of parents that are pure for contrasting traits no form of the trait will appear in the next generation

52:Chromatin fibres undergo folding ,coiling and supercoiling during …….so that the chromosomes become progressively thicker and smaller.

(a)Telophase

(b)Metaphase

(c)Anaphase

(d) Prophase

53:Genetics is the study of

(a) Gene
(b) Chromosomes
(c) Heredity
(d) All

54:Which one is building blocks of chromosomes?

(a) Nucleosome
(b) Chromatin
(c) Chromatid
(d) DNA

55. Genetists or researchers can study the genotype based on the phenotype in classical \forward genetics through…….

(a) DNA sequencing
(b) RNA interference
(c) Gene knockout
(d) Site directed mutagenesis

56:During the summer,the arctic fox produces enzymes that cause its fur to become reddish brown .During the winter ,these enzymes donot function .as a result the fox has a white coat that blends into the snowy back ground .This change in fur colors shows that

(a) The expression of certain genes is affected by temperature
(b) Random alternation of DNA can occur on certain chromosomes
(c) The genes of fur cats are unstable
(d) Mutation occurs

57:According to particulate hypothesis ,parents pass on distinct heritable units .Those distinct heritable units are now called…….

(a) Protein

(b) Genes

(c)DNA

(d)Chromosomes

58:About 90% of the time taken for meiosis is required by

(a)Prophase I

(b)Prophase II

(c)Anaphase I

(d)Anaphase

59:During stage telophase I,chromosomes are…….

(a)Haploid

(b)Diploid

(c)both

(d) polyploidy

60:Mendel determined the segregation of alleles at single gene locus based on

(a) Monohybrid cross
(b) Dihybrid cross
(c) Trihybrid cross
(d) All

61:How much length of the DNA sequence within kinetochores regions?

(a) 120 bp
(b) 140 bp
(c) 160 bp
(d) 180 bp

62:What is the unit of linkage map?

(a) Morgan
(b) Cent-morgan
(c) Centimeter
(d) Angstrom

63:which one of the following statement is not correct about Co-dominance?

(a) A situation ,when both alleles appear in the phenotype

(b) Both allele d in a homozygous inviduals s are expresse
(c) Parallel behivor appears
(d) Both alleles appear

64:Prophase

(a) The nuclear membranes breaks apart and the spindle interacts with the chromosomes
(b) Chromosomes condense and remain attached
(c) Chromosomes duplicates
(d) Chromosomes align in middle of spindle

65:Due to ……. during meiosis I,the two sister chromatids of each chrosome are no longer genetically identical

(a) Crossing over
(b) Chiasmata formation
(c) Metaphase
(d) Synapse

66:DNA replication is a type of ….

(a) Conservative
(b) Semi-conservative
(c) Non –conservative
(d) None

67:Which one of the following statement about prokaryotic chromosomes is wrong?

(a) Prokaryotes have single chromosome plus plasmids
(b) Prokatyotic chromosomes is circular
(c) Prokaryotic chromosomes made of DNA and Histone proteins
(d) Prokaryotic chromosome found in cytoplasm

68:70%genes in human are

(a) Mono allelic
(b) Di allelic
(c) Tri allelic
(d) Tetra allelic

69:Which is the cause of epigenetics?

(a) DNA methylation
(b) Histone modification
(c) mRNA
(d) all

70:Identify the purine base of nucleic acids in the following?

(a) Cytosine
(b) Thymine
(c) Uracil
(d) Adenine

71:The diameter of packed nucleosome is

(a) 2nm
(b) 11nm
(c) 30nm
(d) 300nm

72:In DNA double helix ,the two DNA chains are held together by

(a) Covalent bonds between the pair of bases
(b) Hydrogen bonds between the pair of bases
(c) Ionic bonds between the pair bases
(d) None

73:Which one is not regulate gene expression?

(a)Cytosine methylation

(b)Histone modification

(c)non –coding RNA

(d)Nucleotide medication

74:Epistasis is the interaction between ……genes.

(a)2 (b) 4 (c) 8 (d) 16

75:Technique that gives an rough idea of gene location

(a)Genetic mapping

(b)physical mapping

(c) Top down mapping

(d)bottom up mapping

76:A female bee sex is

(a) XY
(b) Diploid
(c) Haploid
(d) ZW

77:Fragile X chromosomes affect the ..... gene

(a) H19
(b) Igf2
(c) SEC
(d) FMR1

78:Stage of prophase at which nuclear membrane dissolves....

(a) Stage 1
(b) Stage 2
(c) Stage 3
(d) Stage 4

79:The law of independent assortment with dihybrid cross gives genotypic ratio in F2 as.

(a) 1:1
(b) 3:1
(c) 1:2:1
(d) 9:3:3:1

80:Who is the father of genetics?

(a) Charles Darwin

(b) Gregor Johnn Mendel

(c)James Watson

(d)Francis Crick

81:Which of the following is not the property of garden pea plant that was used by Gregor Mendel for his experiment?

(a)Seed in variety of shapes and color

(b)Self and cross pollination

(c)produce many offspring

(d) Long generation time

82:How many numbers of alleles control the fur color of rabbit?

(a) One
(b) Two
(c) Three
(d) Four

83:The ABO blood group in humans is determined by

(a) Multiple alleles
(b) Linked genes
(c) Incomplete dominance
(d) Epistatic genes

84:The giant chromosomes found in …… and they may be as long as 300 micro and 10 micro in diameter.

(a) Diptera
(b) Trillium
(c) Monocot
(d) Dicot

85:Which map is primary based on the location of landmarks along a DNA molecule?

(a) Physical mapping
(b) Radition mapping
(c) Cytogenetic mapping
(d) All

86:Genetic make up of an organism is called

(a) Phenotype
(b) Genotype
(c) Genetics
(d) Trait

87:The type of cell division important for tissue regeneration is

(a) Mitosis
(b) Meiosis
(c) Amitosis
(d) Fission

88:Chromosomes were described by ……in 1875 and chromosome was first used …..in 1888

(a) Gregor mendel,freedrich miescher
(b) Strausberge,waldeyer
(c) Crick,Watson
(d) Thomas,morgan

89:During prophase ,the nuclear envelope……

(a) Disappear
(b) Is intact
(c) Begin to disappear
(d) Forms

90:……..occurs when one mutation has the opposite effect when in the presence of another mutation.

(a) Positive epistasis
(b) Negative epistasis
(c) Sign epistasis
(d) Reciorocal epistasis

91:which of the following has longer chromosomes…..

(a) Plants
(b) Fungi
(c) Algae
(d) Animals

92:A heredity unit chromosomes is called……

(a) Genotype

(b) Gene
(c) Genetics
(d) Trait

93:Which one of the following is not correct about the influence of environmental factors on epigenetics?

(a) The nutrition of the mother can affect the epigenome of fetus
(b) Stress hormones
(c) Social interactions and physical activity not affect the epigenome
(d) Diet affect

94:which of the following statement is correct?

(a) Father with X-linked disorder will transmit the mutant allele to all sons but to no daughters

(b) Father with X-linked disorder will transmit the mutant allele to all daughters but to no sons

(C) Father with X-linked disorder will transmit the mutant allele to no sons but to no daughters

(d) Father with X-linked disorder will transmit the mutant allele to all daughters but to no daughters

95:Name the organism whose first genetic map was made?

(a) Rat

(b) fly

(c)Arabidopsis

(d) Drosophila

96:Which one the statement is not correct about the position dependent method of disease causing gene mapping in human?

(a) Starting from the approximate location of genes to finding itself

(b) Starting at the phenotype ,determining which protein was involved and then getting through the protein

(c) Translation

(d) none

97:Which is the correct sequence of bottom up mapping technique?

(a) make random clones—sequence each clone----overlap sequence reads -----over lap contigs for complete sequence

(b) over lap contings for ----sequencing each colone---overlap

(c)sequence of line---- overlap

(d)over lap timing ----over lap lighting

98:The numbers of genotypes can be calculated on the basis of …….

(a) number of amino acids

(b) Number of alleles

(c)Number of genes

(d)Number of chromosomes

99:When a cell divides.what do we call the 2 new cells?

(a) Mother cells

(b) daughter cells

(c)father cells

(d)son cell

100:which statement is not correct about recessive epistasis in pigeons?

(a) Patteren is epistatics to spread color
(b) Spread patteren
(c) Recessive red
(d) Spread color

101:which is correct about the comb shape in chicken?

(a) Two different comb shapes is controlled by two genes found on two same chromosomes
(b) Four different comb shape is controlled by two genes found on two different chromosomes

(c) Two similar comb are pairless

(d) None

102:chromosomal theory of inheritance states that behavior of chromosome is according to

(a) Mendel law of segregation
(b) Mendel law of independent assortment
(c) Both
(d) None

103:How many hours does the G1 phase of interphase take to complete a cycle?

(a) 4-6 hours
(b) 6-8 hours
(c) 6-12 hours
(d) 16-24 hours

104:In physical map,units of distance are expressed in

(a)Base pairs or Mbs

(b)centimeter or nano meter

(c) Centimorgan or cM

(d)Anstrom

105:Which of the following statement is not correct about Meiosis I?

(a) Meiosis I results in two haploid daughter cells
(b) Reduction division
(c) Homologous chromosome seprate
(d) None

106:A nucleosome consists of

(a) Chromatin and nucleotides
(b) Chromatin and histone
(c) DNA And chromatin
(d) DNA and Histones

107:Which type of inheritance appears in AB blood group phenotype?

(a)Complete dominance

(b) Incomplete dominance

(c) Co-dominance

(d)all

108:The DNA in a bacterial chromosomes is best described as:

(a) A single circular double –helical molecule
(b) A single linear double –helical molecule
(c) A single linear structures
(d) Multiple alleles

109:In humans ,a zygote is a ……..cell having …….chromosomes

(a) Diploid,46
(b) Diploid,23
(c) Haploid,46
(d) Haploid,23

110:Which trait will produce in F1 generation of the crossing a brown pepper (RRyy) with a yellow pepper(rrYY)

(a) Red
(b) Yellow
(c) Brown
(d) Green

111:Which one of the following is an example of diploid cells?

(a) Skin cells
(b) Blood cells
(c) Bone cells
(d) All

112:The splitting of sister chromatids occur at

(a) Propnase
(b) Anaphase
(c) Metaphase

(d) Telophase

113:Which are major factors that affect the epigenome?

(a) Expoure to toxins and diet
(b) Nutrition to mother
(c) Social interaction
(d) Stress hormones

114:Which one of the fur color of rabbit is controlled by maximum numbers of genotypes?

(a) White
(b) Himalayan
(c) Chinchle
(d) Gray

115:Spindle microtubules attach to the kinetochores of chromosomes and move the chromosome to the

(a) Metaphase plate
(b) Telophase
(c) Anaphase plate
(d) None

116:There are about …….loops in the chromosome of E.choli .These loops are highly twisted or supercoiled structure with about …..mlilion nucleotide pairs.

(a) 10,1
(b) 10,2
(c) 25,3
(d) 50,4

117:How many most common chromosomal banding types are present?

(a) 2
(b) 3
(c) 4
(d) 5

118:Giant chromosomes are first discovered by ……in 1881 in dipteran salivary glands.

(a) Balbiani
(b) Watson
(c) Crick

(d) Mendel

119:Pepper color is controlled by

(a) One gene
(b) Two gene
(c) Three gene
(d) Four gene

120:During anaphase of mitosis ……push against each ,that stretches the cell into an oval shape.

(a) Kinetochore spindle fibers
(b) Non –kinetochore spindle fibers
(c) Poles
(d) None

121 ……..provide terminal stability to the chromosomes and ensure its survival.

(a) Telomeres
(b) Centromeres
(c) P –arms
(d) Q- arms

128:Differential expression of the genetic material depending on its parentage of inheritance gives……

(a) Penetrance
(b) Expressivity
(c) Imprinting
(d) Non

129:Which of the following is not the part of interphase in the cell cycle?

(a) S phase
(b) G1 phase
(c) G2 phase
(d) M phase

130:Nuclear DNA replicates in the …… phase.

(a) G1
(b) G2
(c) S phase
(d) M phase

131:Which type of inheritance appears in AB blood group phenotype?

(a) Complete dominance
(b) Incomplete dominance
(c) Co-dominance
(d) All

132:Centromere breaks by the removal of protein …..

(a) Separin
(b) Securin
(c) Heparin
(d) Tubulin

133:Historically ,firstly developed sub discipline of genetics is….

(a) Population genetics
(b) Quantitative genetics
(c) Molecular genetics
(d) Transmission genetics

134:Which of the following statement is correct?

(a) Carrier female mates normal male,100% chance that each daughter will be carrier,50% chance that each son will have the disorder
(b) Carrier female mates normal male,25% chance that each daughter will be carrier,50% chance that each son will have the disorder
(c) Carrier female mates normal male,50% chance that each daughter will be carrier,50% chance that each son will have the disorder
(d) Carrier female mates normal male,50% chance that each daughter will be carrier,50% chance that each son will have the disorder

135:In the parental generation,when Mendel crossed a true breeding pea strain that produced smooth seeds with one that produced wrinkled seeds then result was…..

(a) All the F1 progeny seeds were smooth
(b) All the F1 progeny seeds were wrinkled
(c) half the F1 progeny seeds were smooth

(d)none

136:Which one of the statement is not correct about the position independent method of disease causing gene mapping in human ?

(a) Not knowing anything about the location of genes
(b) Starting at the phenotype ,determining which protein was involved and then getting to the gene through the protein
(c) Forward genetics
(d) Reverse genetics

137:Which of following blood groups are consists of two different genotypes as homozygous and heterozygous?

(a) A&B
(b) A&O
(c) AB&O
(d) B&O

138:Differential expression of the genetic material depending on its parentage of inheritance gives……

(a) Penetrance
(b) Expressivity
(c) Imprinting
(d) Non

139:The molecular weight of the histoneprotein H4 *is*

(a) 21,130
(b) 11,236
(c) 13774
(d) 1333

140:How many numbers of alleles control the fur color of rabbit?

(a) One
(b) Two
(c) Three
(d) Four

141:…….occurs when two deleterious genes are beneficial when together

(a) Positive epistasis
(b) Negative epistasis
(c) Sign epistasis
(d) Reciprocal epistasis

142:In which of the cats the pattern observed in coat color is due to environmental effect?

(a) Tortoise shell
(b) Calico
(c) Siamese
(d) Bob

143:The sugar molecule in a nucleotide is

(a) Pentose
(b) Hexose
(c) Tetrose
(d) Triosde

144:Which one of the following is show the pleiotropic effects in humans?

(a) Sickle cell anemia
(b) Cystic fibrosis
(c) Huntington disease
(d) All

145:When one phenotypic trait is determined by multiple alleles then it is called as…….

(a) Epistasis
(b) Pleiotropic
(c) Polygenic inheritance
(d) All

146:Which of the following is not correct when gene is switched OFF and transcription is not possible

(a) Silent chromatin
(b) Methylated cytosine
(c) Acetylated histones
(d) Deaccelyted histone

147:Chromosome not visible in …….due to ……but can be seen during cell division.

(a) Active nucleus,high water content
(b) Inactive nucleus,low water content

(c) Active nucleus
(d) High water content

148:Histone modification are cause of

(a) Epistasis
(b) Epigenetics
(c) Multiple alleles
(d) Polymorphism

149:More than ……specific trinucleotide repeats is seen in individuals with fragile X

(a) 20
(b) 50
(c) 100
(d) 200

150:Fruit color of summer sqush plant is an example of

(a) Dominant epistasis
(b) Recessive epistasis
(c) Relative epistasis
(d) Multiple allele

151:How many pairs of homologous chromosomes are present in human cell?

(a) 1
(b) 21
(c) 22
(d) 23

153:In life forms with the ZZ-ZW chromosomal sex determination ,the females are

(a) Homogametic
(b) Heterogametic
(c) Microgamatic
(d) Monecia

154:Which one restores the paired condition for both chromosomes and genes

(a) Fertilization
(b) Oogenesis
(c) Spermatogenesis
(d) Mitosis

155:Which of the following organism has highest numbers of chromosomes ?

(a) Human
(b) Dog
(c) Horsetail
(d) Fish

156:Which of the following have single genotype

(a) A&B
(b) AB&B
(c) AB&O
(d) B&O

157:Meiosis,unlike mitosis involves

(a) 1 nuclear division
(b) 2 nuclear division
(c) 3 nuclear division
(d) 4 nuclear division

158:which of the following is not property of Garden pea plant that was used by Gregor Mendel for his experiment?

(a) Variety shape
(b) Color
(c) Long generation time
(d) Many offspring

159:The mitotic apparatus microtubules comprises

(a) Centromeres
(b) Polar microtubules
(c) Asteral microyubules
(d) All

160:……variation describes the situation in which there are many intermediates between the extremes.

(a) Continuous
(b) Discontinuous
(c) Rapid
(d) Random

161:The DNA between nucleosomes is called as linker DNA.its length is ....

(a) 20-80 bp
(b) 10-20 bp
(c) 40-90 bp
(d) 20-60 bp

162:Which one of the following statement is correct about human genes?

(a) 30%genes are di allelic
(b) 30% genes are mono allelic
(c) 30% genes are di allelic and in two forms
(d) None

163:The molecular weight of histone protein H1 is

(a) 22500
(b) 11236
(c) 13774
(d) 13960

164:Genetic material in cells is organized into chromosomes ,prokaryotes generally have……chromosomes.eukaryotes have ….chromosomes in their nuclei.

(a) One circular ,multiple and linear
(b) Two circular
(c) Multiple
(d) Mono

165:Which epistasis between beneficial mutation generates improvements in function?

(a) Positive epistasis
(b) Negative epistasis
(c) Sign epistasis
(d) Reciprocal epistasis

166:Exchange of DNA segments between chromosomes is known as

(a) Multiple alleles
(b) Crossing over
(c) Gene manipulation
(d) Non

167:Which one of the following is not correct about organism and with their chromosomes number ?

(a) Fungi 3-8
(b) Angiosperm-78
(c) Round worm ;2
(d) Protozoa;300

168:The frequency of crossing over between any two linked genes is

(a) Higher
(b) Different
(c) Proportional to distance
(d) None

169:A linkage map

(a) Shows the actual ordering
(b) Spacing of genes on chromosomes
(c) Both
(d) None

170 :G0 phase is '……

(a) Pre –mitotic phase
(b) Post –mitotic phase
(c) Resting phase
(d) Interphase

171:The cytokinesis in animal cells is characterized by

(a) Phragmoplast
(b) Cleavage furrow
(c) Nuclear plate
(d) None

172:Which of the following has longer chromosomes…..

(a) Plants
(b) Fungi
(c) algae
(d) animal

173:Which one of the following genotype expresses the green color of pepper?

(a) RR\YY
(b) Rr\yy

(c) Rr\YY

(d) rr|yy

174:When DNA polymerase is in contact with guanine in the parental strand ,what does it add to the growing daughter strand?

(a) Adenine
(b) Cytosine
(c) Uracil
(d) Thymine

175:When two mutations together have a smaller effect than expected from their effects when alone,it is called

(a) Positive epistasis
(b) Negative epistasis
(c) Sign
(d) None

176:Which one of the following have chromosomes number from 8-16 and most common number is 12?

(a) Fungi
(b) Angiosperms
(c) Round worm
(d) All

177:In case of two gene interaction .the gene which is making the expression of another is called……

(a) Dominant
(b) Recessive
(c) Epistatic
(d) Non

178:DNA replication is type of …..

(a) Conservative
(b) Semi-conservative
(c) Non-conservative
(d) All

179:Hair color is controlled by alleles on ……

(a) Chromosomes 2,4,8,10
(b) Chromosomes 3,6,10,18

(c) Chromosomes 4,6,10

(d) Chromosomes 5,7,8

180:Groups of human chromosomes are…..

(a) 3

(b) 5

(c) 7

(d) 9

181:During the summer ,the arctic fox produces enzymes that cause its fur to become reddish brown,During the winter ,these enzymes do not function .As a result,the fox has white coat that blends into the snowy background.this change in fur color shows that

(a) The expression of certain genes is affected by temperature

(b) Random alteration of DNA

(c) Fox is missing

(d) Mutation

182:All offspring of same variety is called as

(a) Cross breed

(b) True breed

(c) Line breed

(d) Hybrid

183:Which one of the following statement is not correct about incomplete dominance?

(a) Intermediate phenotype appear

(b) One of the allele is dominant

(c) Phenotype appears as blend of both allele

(d) Neither allele is dominant

184:Stage of prophase at which nuclear membrane dissolve…

(a) Stage 1

(b) Stage 2

(c) Stage3

(d) Stage4

185:In meta phase I,……line up at metaphase plate with one chromosome facing each pole

(a) Triads

(b) Tetrads

(c) Pentads

(d) Hexads

186:Human chromosomes are

(a) 22 pair homologous
(b) One pair non homologous
(c) Both
(d) None

187:The diameter of chromatin is

(a) 2nm
(b) 11nm
(c) 30nm
(d) 300nm

188:The life cycle consists on

(a) Haploid egg+haploid sperm_diploid zygote_mitosis
(b) Diploid adult
(c) Meiosis+haploid gametes
(d) All

189:Building block of nucleic acids are…..

(a) Histones
(b) Amino acids
(c) Nucleotides
(d) None

190:……,states that the two alleles of gene found on each of pair of chromosomes separate independently of one another into sex cell

(a) Independent assortment
(b) Segregation
(c) Random inheritance
(d) All

191:E.choli has about…….. genes are organized into its one circular chromosome

(a) 1000-2000
(b) 2000-3000
(c) 3000-4000
(d) 4000-5000

192:The genome length of prokaryotic chromosome is smallest in

(a) RNA virus
(b) DNA
(c) Both
(d) None

193:Which one of the following statement is correct about variation?

(a) Influence of genetic makeup
(b) Environmental factor
(c) Both
(d) None

194:Spindle microtubules attach to the kinetochores of chromosomes and move the chromosomes to the

(a) Metaphase plate
(b) Telophase
(c) Anaphase
(d) None

195:Modern concept about the law of dominance is……

(a) The gene for a particular inherited character resides at specific locus homologous chromosome
(b) The gene for a particular inherited character resides at specific locus on heterogenous chromosomes
(c) Both
(d) None

196:Technique that gives an rough idea of gene location

(a) genetic mapping
(b) physical mapping
(c) top –down
(d) none

197:In blue_eyed Mary plant a white substance is first converted to ……..substance by 1st gene and that to a ……substance by 2nd gene to give a blue flower

(a) colorless,magenta
(b) blue,red
(c) violet
(d) green

198:.....variation describes the situation in which there are many intermediates between the extreme

(a) continous
(b) rapid
(c) random
(d) none

199:.....cells become .....cells of the embryo which differentiate into specific lineage

(a) totipotent,pluripotetn
(b) rando ,earm
(c) totipotent
(d) plurpotent

200:Which condition is called as polygenic inheritance?

(a) Influence one trait
(b) Multiple trait
(c) Two genes influence one trait
(d) Ditrait

201:How many number of bands can be recognized on giant chromosomes of the Drosophila?

(a) 50
(b) 500
(c) 5000
(d) 50000

202:What is the other name of polytene chromosomes?

(a) Salivary gland chromosomes
(b) Lamprush chromosomes
(c) Regulatory chromosome
(d) None

203:In the ABO system,blood group O is characterized by the

(a) Absence of both antigen A and antigen B
(b) Absence of A
(c) Presence of antigen O
(d) None

204:Shepherds purse has two shape heart and narrow .If you cross a true breeding to heart to a true narrow .F1 will give….

(a) Heart
(b) Narrow
(c) Disc shaped
(d) None

205:what is the name of the region where double stranded singular circular DNA is found in the prokaryotic cell?

(a) Protonucleus
(b) Nucleus
(c) Nucleoid
(d) Nucleoplasm

206:What is the composition of nucleotides?

(a) A base +a sugar +phospholipids
(b) A base
(c) A sugar
(d) None

207:Which of the following is widest DNA?

(a) A-DNA
(b) B-DNA
(c) C_DNA
(d) Z_DNA

208:Causes genes to not segregate independently ,causes genes to be more likely to be inherited together

(a) Gene linkage
(b) Epistasis
(c) Polygenic
(d) All

209:It is estimated that about ……genes are imprinted and they can be found on several different chromosomes.

(a) 20

(b) 40
(c) 60
(d) 80

210:Tetrads align at center ….

(a) Prophase I
(b) Metaphase I
(c) Anaphase I
(d) None

211:Meiosis in males is called ……and produces sperms

(a) Ovulation
(b) Fertilization
(c) Spermatogenesis
(d) Division

212:Primary occyte produces polar bodies….

(a) 1
(b) 2
(c) 3
(d) 4

213:In humans ……of somatic cells

(a) ~20 types
(b) ~120 types
(c) ~220 types
(d) ~420 types

214:Each identical side of a chromosomes is the

(a) Chromatid
(b) Centriole
(c) Chromatin
(d) Centromere

215 : How many types of enzymes are involved in histone acetylation?

(a) 2
(b) 4
(c) 6

(d) 8

216:The region where two sister chromatids of a chromosome joined or held together during mitoitic metaphase is called…

(a) Telomeres
(b) Centromere
(c) P arms
(d) Q arms

217:Which trait will produce inF1 generation of the crossing over a brown pepper (RRyy) with yellow pepper (rrYY)?

(a) Red
(b) Blue
(c) Violet
(d) Green

218:In life forms with the ZZ-ZW chromosomal sex determination ,the males have

(a) ZZ chromosomes
(b) WW chromosomes
(c) Both
(d) None

219:which one of the following statement is correct about to the Mendel Law Of Dominance ?

(a) In a cross of parents that are pure for contrasting traits only one form of the trait will appear in the next generation
(b) a cross of parents that are pure for contrasting traits only one form of thetrait will appear in the next generation

(c)cross of parents that are pure for contrasting traits only one form of the trait will appear in the next generation

(d) none

220:A point mutation in the …….completely eliminates the ability of the centromere to function during chromosome segregation

(a) CDE-I sub domain of CEN DNA
(b) CDE-II sub domain of CEN DNA
(c) CDE- III sub domain of CEN DNA
(d) None

221:Which one of the following statement is not correct about Prophase II?

(a) Replication of chromosomes occur
(b) Nuclear membrane disappears
(c) Spindle fibers forms
(d) none

222:Which statement is not correct about epigenetic regulation in identical twins?

(a) The epigenetic tags can have such an effect on the twins that one can develop a disease and other is also
(b) Difference twins are form
(c) Over time twins form
(d) All

223:Which one of following organisms have 16 chromosomes ?

(a) Onion and mold
(b) Rice
(c) Tomato
(d) All

224:T.H Morgan test cross …..

(a) Genes located on same chromosomes
(b) Crossing over and linkage genes that breaks the gene physical connection
(c) Body change
(d) All

225:Which of following does not revert to the euchromatin stage?

(a) Constitutive heterochromatin
(b) Constitutive euchromatin
(c) Facultative heterochromatin
(d) All

226:Which one of following statement about color blindness is correct?

(a) Color blind mother will transmit the mutant alleles to all sons but to no daughter
(b) Color blind mother will transmit 50%
(c) color blind will transmit 45%
(d) all

227:Which of following statement is correct about the epigenetic modification?

(a) When histones are acetylated chromatin is open and genes are potentially active
(b) Histones not tagged
(c) Histones tagged
(d) All

228:How many numbers of heterozygotes genotypes will be produced from five number of gene specific alleles?

(a) 3
(b) 4
(c) 15
(d) 60

229:Genes will be switched ON when

(a) Histones are acetylated
(b) Histones are deacetylated
(c) Cytosine
(d) Methionine

230:Which one is first time observed by Flemming in 1882?

(a) Salivary gland chromosomes
(b) Lampbrush chromosomes
(c) Regulatory
(d) Giant

231:Chargaffs rules for DNA base pairing indicates that

(a) A=C
(b) A+T=G+C
(c) D=C
(d) A=T

232:Failure in which of the following phenomenon would result in linkage?

(a) Law of separation
(b) Law of independent assortment
(c) Law of segregation
(d) All

233:Cell splits into

(a) Interphase
(b) Metaphase
(c) Telophase
(d) Cytokinesis

234:Skin color pigmentation in humans is controlled by at least …..separately inherited genes?

(a) 1
(b) 2
(c) 3
(d) 4

235:Formation of pluripotent cells from totipotent cells is called

(a) Genomic imprinting
(b) Cellular differentiation
(c) X chromosomes
(d) Formation

236:What type of segregation occurs when alleles at the two gene loci segregate together and are transmitted unit?

(a) Dependent segregation
(b) Independent segregation
(c) Both
(d) None

237:Fragile X syndromes is most commonly caused by

(a) ***238:Which one of*** *CG*
(b) GG
(c) CGG
(d) GGG

the following statement is not correct about homologous chromosomes

(a) Move to same location
(b) Similar in shape
(c) Similar in size
(d) All

239:The tendency of two or more than two genes to stay together during inheritance is called

(a) Genetics
(b) Gene interaction
(c) Crossing over
(d) Linkage

240:How many numbers of alleles can generates 3 genotypes?

(a) 2
(b) 5
(c) 3
(d) 4

241:Group of adjacent nucleotides are joined by

(a) Phosphodiester bond
(b) Peptide bond
(c) Ionic bond
(d) Covalent bond

242:Which is correct about the comb shape in chickens?

(a) Four different comb
(b) Two different comb
(c) Five different comb
(d) All

243:…….occurs when one mutation has the opposite effect when in the presence of another mutation.

(a) Positive epistasis
(b) Negative epistasis
(c) Sign epistasis
(d) All

244:A cell with 10 chromosomes undergoes mitosis .How many daughter cells are created?…..Each daughter cell has…..chromosomes.

(a) 10,2
(b) 2,10
(c) 1,10
(d) 2,10

245:The cross –over frequency between the two genes is

(a) 7%

(b) 5%
(c) 2%
(d) 50%

246:which one is not regulate gene expression?

(a) Cytosine
(b) Histones
(c) Non coding
(d) Nucleotide modification

247:During recessive epistasis ,……,gene exerts its effectwith …..genotype

(a) Epistatic,homozygous,recessive
(b) Dominat ,hetero
(c) Recessive
(d) All

248:Name of the organisms whose first genetic map was made?

(a) Rat
(b) Fly
(c) Drosophila
(d) All

249:histone modification is cause of

(a) Epistasis
(b) Epigenetics
(c) Multiple alles
(d) Poly morphism

250:Which statement is not correct about recessive epistasis in pigeons?

(a) To spread color
(b) Recessive red
(c) Both
(d) None

251:Haploid gametes are formed by

(a) Mitosis
(b) Meiosis
(c) Amitosis

(d) Fission

252:How many numbers of genes are present in yeast?

(a) 2000
(b) 5000
(c) 5800
(d) 1300

253: Nuclear DNA replicates in the ……phase.

(a) G1
(b) G2
(c) S phase
(d) M phase

254:The type of cell division important for growth is

(a) Mitosis
(b) Meiosis
(c) Amitosis
(d) Fission

255:Kinetochore are formed at

(a) Telomeres
(b) Sis chromatids
(c) Centromeres
(d) All

256:which stage of the cell cycle are arrested before staining?

(a) Prophase
(b) Anaphase
(c) Metaphase
(d) None

257:which type of genetics deals with the movement of genes traits

(a) Population
(b) Transmission
(c) Molecular
(d) All

258:The chromosomes number can be counted easily during….

(a) Mitotic metaphase
(b) Anaphase
(c) Telophase
(d) All

259:The pleiotropic effect is shown in human

(a) Sickle cell anemia
(b) Cystic fibrosis
(c) Huntigton
(d) All

260:The molecular weight of histone protein is

(a) 22500
(b) 222000
(c) 2222
(d) 22223

261: The diameter of DNA double helix is

(a) 2nm
(b) 30nm
(c) 5nm
(d) 6nm

262:In which of the cats the pattern observed in coat color is due to environmental effect?

(a) Tortoise shell
(b) Calico
(c) Siamese
(d) Bob

263:According to …….genetic material of the parents mixes during fertilization

(a) Particular hypothesis
(b) Blending hypothesis
(c) Both
(d) None

264:The correct sequence is ….

(a) Lipotene ,zygotene
(b) Pachytene,diplotene
(c) Diakinesis
(d) All

265:when a cell divides ,what we call 2 new cells?

(a) Mother cells
(b) Father cells
(c) Daughter cells
(d) Son cells

266:A DNA polymorphism is …..of population

(a) 1-2%
(b) 2-3%
(c) 4-6%
(d) 6_7%

267:Generally ,karyotypes represent by arranging the …….. order of size keeping their centromere in a ….

(a) Ascending,vertical lines
(b) Descending ,straight line
(c) Ascending ,straight line
(d) All

268: when ice pack is attach to Himalayas rabbit ….

(a) Color will be darker
(b) Die
(c) Degenerates
(d) All

269:What is phosphodiesters linkage

(a) 5 phosphate group
(b) 3 hydroxyl group
(c) Both \
(d) All

270:The sugar molecule in nucleotides is

(a) Pentose

(b) Xexose
(c) Hexose
(d) Triose

271:The two duplicated homologous separate during the anaphase stage ….

(a) Meiosis i
(b) Mitosis
(c) Anaphase
(d) All

272:The diameter of condensed scaffold associated chromosome is

(a) 11nm
(b) 29nm
(c) 300nm
(d) 700nm

1----------Which of the following is true about B-DNA?

planes of the base pairs nearly perpendicular to the helix

2-------Which one of the lowing statement is conect about X-chromosome-tvstion phenomena?

One expression on one of the male X-chromoson is down regulated due to DNA methyletion an O

3-----Darker bands are generally found near the centromere or on the ends (telomeres) of

Hetrochomatic region heterochromatin

4----------Genetics is the study of

All of the mentioned

5---------- Wachu dicipline is called on Mondulien genetics?

Quantitative genetics

6----------: Epestesis is the interaction between

7----------Exchange of DNA segments between chromosomes is known as

Chiasmata

8---------: Which one of the following statement about eukaryotic chromosome is wrong?

Eukaryotic chromosomes found in a cytoplas

(a) Lipotene ,zygotene ,

[9----------The correct sequence of Meices I prophans stages is?

Leptotene, zygotene pachytore diplotne danes

10---------- Which of the following enzymes remove helix and produces sick to

Dyrses and Topoisome

11----------10-15% of chromosomal deletions occur due to

5000

12----------Chromosomes having centromere at one end of arms

Telocentric

13====== The agents which induce artificial mutations are called

Mutagenic agent

14----------In case of summer squash, which statement is not correct?

Dominant Y locus produces yellow color

15----------During ..,.....push against each other, that

Non ko.....

16----------

Which one of following allele controls the Chinchilla (ight gray) color of the robbit?c

C

17----------The albitic series refers to

Lethal alickes

18----------: Inherited variations genetically .....and cannot be altered

Controlled

19----------: What enzyme is responsible for copying methyletice marks fces the per tot of DNA during replication?

DnmTg

20----------: .....Causes a trait to appear in inappropriate tissues or at inappropriate times during the process of development

Grain of function

25----------: In physical map. units of distance are expressed

Centmorgan of Cm

270:The sugar molecule in nucleotides is

26---------- The diameter of DNA double helix is

(a) Pentose

11nm

27----------Which one of the following genotype expresses the Green color of pepper?

rrlyy

(b) Xexose

28----------: A nucleosome consists of

Dna and histones

29----------: Which one of the following statement is not correct about Co-dominance?

Both alleles are expressed in a heterozygous individuals

30----------Which of the following is not correct when gene is switched OFF and transcription is not

Acteylated histones

31----------: E coli has about .....genes are organized into its one circular chromosome

3000to 4000

32----------Which one of following statement is not correct about Homologous Chromosome

Pair of chromosomes that are similar in shape

33…….Which of the following is not a common mode of histone modification in eukaryotes?

Sulphonation

34----------Band that indicates constitutive heterochromatin

G band

35----------A female bee sex is

Dipkid

36----------: Most common phenotype in natural populations of the organism is called

Wild type phenotype

37----------: How many numbers of heterozygotes genotypes will be produced from five numbers of

Ten

38----------In Me forms with the ZZ-ZW chromosomal sex determination, the males are

Homogametic

41----------: Which of the following is true about B-DNA?

Planes of the base pairs nearly perpendicular to the hela exis

43----------Which of following statement is correct about the epigenetic modification?

When histones not tagged deacetylated the chromatin open and genes active

[44----------Which one of following statement about color blindness is correct?

Color blind mother will transmit the mutant allele to 50% sons but to 50% daughters

45---…….alters the amino acid sequence of the protein but does not change to function replaced amino acid is chemically similor or the affected amino and has lelo luence on pi function

Neutral mutation

(a) Lipotene ,zygotene ,

46----------: The type of cell division important for growth is

Mitosis

47----------Suppose that the distance between two genes on a linkage mep is 7 units. The crom over fruquency between the two genes is

7%

48----------: Which one of the following is not correct about organism with their chromosome numbers?

Angiosperms have 78 chromosomes

49----------: Which one of tollowing statement is not comect about Mis

Meiosis I results in two haploid daughter cells

50----------: Name the phenomenon which can break the occurrence of linkage?

Crossing over

51----------: During metaphase mitosis chromosome

Lineup at the equator

52----------Hetorically, firstly developed sub discipline of genetics is

Transmission genetics

53----------Which statement is not correct about epigenetic regulation in identical twins?

The epigenetic tags can have such an effect on the twins that one can develop a disease and the other is also

54----------: How many hours does the G1 phase of interphase take to complete a cycle?

6-12hours

55----------: How many numbers of alleles can generale 10 genotypes?

Genes with 4 alletes

56----------: In ife forms with the ZZ-ZW chromosomal sex determination, the males have

zz checmasone

57----------: Chargaff's rules for DNA base pairing indicates that

A-T and G-C

270:The sugar molecule in nucleotides is

58----------Dicots in general, have a.... number of chromosome than monocots. Chromosomes a..... in monocot than dicots

(a) Pentose

Higher longer

(b) Xexose

59-----Mitotic apparatus helps

Chromosome inheritance

60------Which of the following produces monosomic and trisomic daughter cells Select the

Meiotic I non-disjunction

61--------[: When chromosomes stained they show a dark-stained region that is the

Centromere

[65----------What will be dihybrid test cross ratio if genes are located on different chromo

933

66-------- Which one of following organisms' series have no metacentric autosomal chromosome

Dog.goat.cow

[67---------- Historically, firstly developed sub discipline of genetics is

Transmission genetics

[68----------What is the division of cytoplasm colled?

Telophane

[69----------: During anaphoon of loss push against each other that stretches the call

Non kinetochore sprdietbers

70---------- Which of the following blood group es considered a universal donor?

B

71---------- Which of the following is not correct when gene is switched ON and transcription is possib

Acetyloted histone

[72---------- The region where two sister chromatids of a chromosome joined or "held together"

Parms

73---------- Fegle X chromosome affects the gene

Fmr1

[74: - ----------How many numbers of homozygotes genotypes will be produced from

4

[75----------: In which of the following fur color of the rabbit is controlled by

Chinchilla(light gray)

[76----------: ich genotype is not of the golden cost color in Labrador retriever?

bbEE

77---------- Building blocks of nucleic acida are

Nucleotides

78---------- The numbers of genotypes can be calculeted on the basis of

Numbers of oteles

79----------70% genes in human are

Mono-allelic

80----------The giant chromosomes found in and they may be as long as 300 μ and up to

Diptera

81---------- How many number of bands can be recognized

5000

82----------: two duplicated homologues separate during the an

Meiosis I

83---------- Generally, karyotypes represent by arranging in the

Descending straight line

84---------- Transmission of traits from parents to offspring

Heredit

85---------- The nuclieone is composed of a core of histone proteins and

Six 160

86----------: Whose exp.. result gives the the first solid evidence indicating

T h.morgen

87---------- Which of the following is not correct when gone is switched ON and transcription

Acetylated histones

88----------: how many numbers of Allen can generale 10 genotypes?

Genes with 4 alleles